Building the Robotic Arm

Summary:

This lesson gives students hands-on experience physically assembling the robotic arm they will use for the remainder of the course. Students work in groups to construct the arm's links and joints, wire the actuators to the controller, and perform an initial functional test to confirm every joint moves correctly. The lesson also introduces basic hardware troubleshooting, giving students practice identifying and fixing common assembly issues before moving on to the kinematics content in later lessons.

Objective:

By the end of this lesson, students will be able to:
● Assemble the mechanical components of the robotic arm (links, joints, mounting hardware) following a build guide
● Correctly wire actuators/servos to the controller
● Verify that each joint moves correctly through basic functional testing
● Identify and troubleshoot common assembly errors (loose connections, reversed wiring, misaligned joints)

Standards:

NGSS

Resources & Materials:

Arm Material Kit
Assembly Instructions

Procedures:

Day 1:
Tool safety reminder (5 min) — proper use of screwdrivers/hex keys, general workstation safety
Build overview (10 min) — walk through the build guide at a high level, preview the order of operations
Assembly, part 1 (40 min) — students work in groups assembling links and joints per the build guide
Wrap-up & cleanup (5 min) — organize remaining parts, note progress for next class

Day 2:
Recap (5 min) — review assembly progress from Day 1
Wiring (25 min) — connect actuators/servos to the controller per the wiring diagram
Initial power-on test (15 min) — power the arm, verify each joint responds correctly
Troubleshooting (10 min) — instructor-guided fixes for any joints that don't move correctly (reversed wiring, loose connections)
Wrap-up (5 min) — confirm each group's arm is fully functional before moving to FK lessons

Assessment:

1. Why is it important to follow the build guide in order?
a) It looks neater
b) Some parts are harder or impossible to attach later if skipped
c) It's required by the manufacturer's warranty
d) It doesn't matter, order is just a suggestion

2. If a joint doesn't move after power-on, what should you check first?
a) The link's color
b) The end effector
c) The wiring connection
d) The workspace

3. What is one purpose of the wiring diagram during assembly?
a) To decide what color the arm should be
b) To measure the arm's workspace
c) To calculate the arm's degrees of freedom
d) To correctly determine which wires go to connections

4. What do you do if an assembly is built incorrectly?
a) Force the remaining steps to continue as planned
b) Stop, disassemble back to the point of the error, and rebuild correctly before continuing
c) Leave it as is and note it for later
d) Add extra fasteners to compensate

Answer Key

1. B
2. C
3. D
4. B